

Constrained, not identifiable: statistical constraints on the production of the Voynich manuscript text

Sebastian Beyer

Independent researcher, Vienna, Austria. Correspondence: @BeyerSebastian on X.

Abstract. Which production mechanisms are compatible with the text of the Voynich manuscript, and can text statistics decide whether its free choices carry a message? An exploratory analysis shows that ordering information between tokens sits in the glyphs at the token boundary, that line breaks interrupt it, that real tokens are attested no more often than tokens sampled from a glyph model, and that the text carries no measurable information about the colours of its drawings. A low-order glyph process with drifting parameters reproduces these statistics on held-out pages; published generators and the Naibbe cipher do not. We then applied a battery of calibrated tests, specified before the Voynich outcomes were computed. Its controls are handwritten human gibberish, natural languages, historical-style ciphers and generators with known mechanisms. Seven further pre-specified experiments followed. Nine measured properties constrain the mechanism, among them rigid slot order within words, margin-associated line-final forms, short-range recency across line and paragraph boundaries, and very low recurrence of locally deviant forms. They exclude naive improvisation, stationary procedures, settings that switch per line or paragraph, copy-and-modify with propagating innovations, the tested natural-language controls, the Naibbe cipher as published, and copying from a re-lined draft. The best explicit generator reproduces 33 of 40 statistics on held-out pages. Finally, arithmetic decoding lets the free choices of a Voynich-like process carry about 44 kB of compressed plaintext per manuscript, recovered without error, with no statistic changing beyond run-to-run variation in 20 runs per condition, also when the message drives the best explicit generator. The production mechanism is constrained, but whether its free choices carry a message is not identifiable from the text alone.

Keywords: Voynich manuscript; mutual information; mechanism identification; human gibberish; Naibbe cipher; mimic functions; non-identifiability

1. Introduction

The Voynich manuscript is a codex on vellum radiocarbon-dated to the early fifteenth century (Zandbergen n.d.). It contains about 38,000 word tokens in an unidentified script, written by several scribes (Currier 1976; Davis 2020). A century of decipherment attempts has not produced an accepted reading (D'Imperio 1978; Bower and Lindemann 2021). Quantitative work has described its unusual properties: two statistical "languages" (Currier 1976), low conditional glyph entropy and a rigid word structure (Stolfi 2000; Lindemann and Bower 2020; Zattera 2022), strong coupling between the last glyph of one word and the first of the next (Rozanova and Temerev 2026), and a positional entropy profile shared with some verbose ciphers (Kinnison 2026). Constructive proposals have shown that such properties can arise from a table and grille (Rugg 2004), from copying and modifying earlier words (Timm and Schinner 2020), from a historically executable cipher (Greshko 2025), or from human gibberish (Gaskell and Bower 2022). Bower and Gaskell (2022) showed that several encipherment methods, among them encoding letters as bigraphs, merging phonemic contrasts and adding nulls, bring character-level statistics of Latin close to Voynich values, although no single method matched on all of their metrics. Hermes (2022) showed that the Polygraphia III of Johannes Trithemius (1506), which replaces each plaintext letter with an invented word, approaches the Voynich difference between unigram and conditional glyph entropy, its word-length distributions and its many near-identical neighbouring words when the substitution columns are chosen at random.

Results of this kind establish sufficiency: a mechanism *can* produce the text. They do not establish identity, because the same statistics are compatible with a procedure followed by rule, with a practised writer's habits, and with a cipher whose free choices are driven by a message. This paper

replaces the question "which mechanism fits?" with two others. First, which mechanisms does the text exclude, under tests whose ability to discriminate has been shown on controls with known mechanisms? Second, is there any statistical route from the text alone to the question of whether a message is present?

We answer both in two parts. The first part (Section 3) is exploratory. It applies one battery of information-theoretic statistics, with marginal-preserving null models, to the manuscript, to language controls and to candidate generators, and shows that a local glyph process suffices. It was not specified in advance, and it shaped the hypotheses of the second part. In the second part every analysis was specified in a written plan, committed to the project repository before its Voynich outcome was computed (Section 4.3). A battery of seven tests (T1–T7) was calibrated on controls first and then applied to the manuscript. Seven further experiments (E1–E7) were designed to separate the mechanisms the battery left standing, to build explicit generators of the surviving class and test them on held-out pages, and to test constructively whether a message can hide in the free choices.

The contributions are:

1. An exploratory profile of the text: ordering information carried by the glyphs at token boundaries rather than by whole tokens, a lexicon test in which real tokens are attested no more often than tokens sampled from a glyph model, a sufficient local glyph process tested on held-out pages, the characteristic failures of published generators and of the Naibbe cipher, and a text–illustration test with calibrated sensitivity.
2. A two-axis hypothesis-separation framework: generation by fixed procedure, skilled improvisation or plaintext-driven procedure, crossed with direct composition or copying. Every discriminator has predictions and falsifiers stated in advance and a calibration requirement on controls.
3. A mechanism class defined by nine measured properties, each with its evidence and an explicit strength label (Table 8). Several are new empirical findings: line-final forms associated with the margin and much weaker at paragraph ends, recency that runs continuously across line and paragraph boundaries, very low recurrence of locally deviant forms, page clustering even of the most frequent forms, and within-page drift that is lexical in type.
4. Exclusions of named mechanism variants, kept separate from the alternatives that remain open (Table 9).
5. Held-out tests of explicit generators with ablations, which identify necessary components and the properties no tested generator reproduces.
6. A constructive non-identifiability result for this text. The general principle, that a message can drive the random choices of a known process without changing its output distribution, is established (Wayner 1992; Cachin 2004; Hopper, Langford and von Ahn 2002). What is new is its instantiation for a Voynich-like process on the manuscript's layout, the measured capacity of that process (1.90 bits per glyph), and its evaluation against the full 40-statistic phenotype, with 20 runs per condition and for two processes. Text statistics therefore cannot settle the plaintext question without further assumptions about the encoder.

The strongest conclusion the evidence supports is this: **a set of production constraints can be recovered from the text, but message-carrying and message-free processes of the constrained class are not identifiable from the text alone.** Nothing in this paper shows that the Voynich text is meaningless, and nothing decodes it.

2. Materials

2.1 Transliterations and units

The primary transliteration is ZL3b (Zandbergen and Landini 2025) in the IVTFF format. We use paragraph text (loci of type P), join uncertain spaces (commas; the alternative is a robustness check), and drop tokens containing unreadable or rare-glyph codes. This leaves 32,428 tokens on 207 pages. In the exploratory profile (Section 3) a glyph is a single EVA character. The pre-specified tests segment glyphs into units, with the multigraphs *cth*, *ckh*, *cph*, *cfh*, *ch*, *sh* as single units. The Takahashi transliteration IT2a (34,411 tokens) is aligned by page and line. A token is *double-coded* when both transliterations read it identically in lines with equal token counts. Page metadata give the quire, the illustration section, the Currier language and the scribal hand, following Davis (2020). A line is paragraph-final if it carries the paragraph-end marker, otherwise margin-bound. Page order is the current binding order.

2.2 Controls

Every test is calibrated on corpora whose mechanism is known.

- **Human gibberish:** the 38 handwritten documents collected by Gaskell and Bown (2022), 10,075 tokens in lowercase Latin letters with the writers' line breaks preserved. Each writer invented a different "language".
- **Natural language:** Caesar's *De bello Gallico* (Latin), Dante's *Commedia* (Italian), Goethe's *Faust* (German) and Tolstoy's *War and Peace* in English, cut into the Voynich page and line shape. The profile adds consonantal Hebrew (Genesis) as an abjad with short words. Latin is also wrapped greedily at a fixed width, as a re-lined text for the line tests.
- **Ciphers:** the Naibbe cipher (Greshko 2025), reimplemented from the published tables, checked against the author's own published ciphertext (Section 3.3), and enciphering Latin as running text and as independent lines, with state-dependent variants; a verbose substitution with deterministic homophones; and ciphertext-feedback substitution.
- **Synthetic generators:**
 - an order-3 glyph automaton per language and section, with and without page drift;
 - a table and grille after Rugg (2004);
 - copy-and-modify after Timm and Schinner (2020);
 - a periodic device with column and row effects of adjustable strength;
 - generators with smooth or stepwise parameter drift, settings per line or paragraph, explicit recency, and vocabulary or glyph-habit drift.

Generators fitted to or built from Voynich statistics are used only to show how a test responds to a known mechanism, never as evidence of fit.

2.3 Illustration features

For the text–illustration test we computed ten colour features of each page from the Beinecke digital facsimile of folios 1r–58v: the area shares, centroids and hue of green, blue and red–brown pixels, classified relative to the page's parchment colour (Supplement S1). They are crude proxies for leaf mass, flowers and roots, chosen to test one hypothesis: that the text of a herbal page describes visible properties of its drawing.

2.4 Held-out split

For generator tests the manuscript is split by the parity of the bifolio index within each quire. Conjugate leaves stay together, and every quire and the three main hands occur in both halves. Generators see only the training half's text, plus the layout and metadata of test pages (line lengths,

paragraph ends, quire, section, language).

3. An exploratory profile of the text

Before the pre-specified program we applied one battery of statistics to the manuscript, to the language controls and to candidate generators. This analysis was exploratory: it was not specified in advance, and its results shaped the hypotheses of Section 4. It establishes the properties the later tests build on. Its main result, that a local glyph process suffices, is the starting point for asking what the text excludes. Methods are summarised here and given in full in Supplement S1. All tables of this section are in Supplement S2.

3.1 Measurements

A token is a string between spaces; inner tokens are neither first nor last in their line. Lines with fewer than two tokens are dropped, which leaves 32,399 tokens. Mutual information (MI) is estimated by plug-in and is interpreted only as the excess over a null model that preserves the marginals exactly, because the plug-in bias depends on sample size and marginals (Paninski 2003). Order within lines is tested against a permutation of the inner tokens of each line. Pairs that span lines, and pairs of token and page, are tested against a re-pairing null. Corpora are compared at equal numbers of pairs, and the controls are cut into the line, paragraph and page lengths of the Voynich text. Uncertainty is a leave-one-quire-out jackknife over 16 quires.

3.2 Five properties

Ordering information sits at the token boundary (Figure 1a; Table S2). Adjacent inner tokens share 0.076 bit of excess MI as whole tokens, but the last glyph of one and the first glyph of the next share 0.146 bit. In the languages the order is reversed. Whole tokens share 0.178 (Hebrew) to 0.409 bit (Italian), and the boundary glyphs 0.037 (Latin) to 0.152 bit (Italian, whose elisions couple word edges). A projection of a pair cannot carry more information than the pair, so the token-level estimate is attenuated by sparsity. The finding concerns where the information is carried, not how much there is. The coupling is stronger in Currier B (0.187) than in Currier A (0.082) and agrees with Rozanova and Temerev (2026).

The excess lies in pairs of similar tokens, not in collocations of dissimilar words. Pairs at edit distance 3 or more carry 0.243 of Latin's 0.262 bit but -0.075 of the Voynich text's 0.203 bit (Table S23). This similarity of neighbours reflects the page, not copying from the line above. The closest of the three tokens above a token lies within edit distance 1 as often as the closest of three random tokens of the page (0.108 against 0.109; Table S22).

Line breaks interrupt the coupling (Table S3). The last token of a line and the first token of the next share 0.005 bit, against 0.062 ± 0.007 for adjacent pairs at the same n . At the glyph level the values are 0.012 and 0.144. In the language controls the two are equal (Latin 0.077 and 0.070), and even Latin written as one independent sentence per line keeps 0.159 bit across the break. The page-conditional estimator of Section 5 (T1), the only one we found valid under writer heterogeneity, puts the ratio at 0.22 with a standard error of 0.25. The coupling is therefore strongly reduced at line breaks. Whether it is absent is not decided.

Token boundaries are predictable from the flanking glyphs (Table S4). One glyph on each side gives an information gain of 73.7% about whether a boundary is present ($F_1 = 0.85$), against 38.7% to 44.9% in the languages. Latin stripped of its spaces and re-spaced by a fixed glyph rule gives 73.2%.

There is no measurable lexicon (Figure 1b; Table S5). A within-token glyph trigram model is fitted on a random half of the tokens. Tokens sampled from it are attested in that half as often as held-out real tokens (0.76 against 0.79), and the jackknife interval of the gap includes zero. In every language, and in Latin cut into rule-spaced pieces or syllable pairs, sampled tokens are

attested far less often (Latin 0.20 against 0.81; rule-spaced Latin 0.28 against 0.62). As far as this test can tell, a Voynich token is whatever the local glyph statistics produce.

The entropy profile is that of a low-order automaton (Table S6). With four symbols of context, the conditional entropy of the glyph stream is only -0.06 bit below that of an order-3 twin of the same size. The languages lie -0.30 (Latin) to -0.13 bit (Hebrew) below theirs. Under lzma compression, which sees all ranges, the residual of the text against a twin whose order-3 parameters drift by section is -0.039 bit per glyph. Samples of the automaton itself give $+0.033$ and -0.030 ; Latin gives -0.397 .

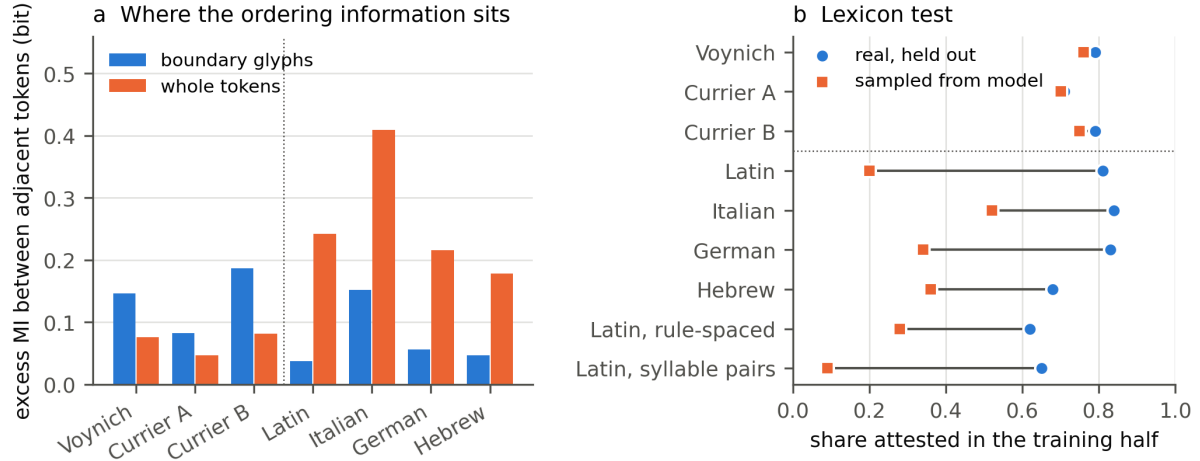


Figure 1. (a) Excess MI between adjacent inner tokens, for the boundary glyphs (last glyph of a token, first glyph of the next) and for whole tokens, against within-line permutation. (b) Lexicon test: share of held-out real tokens and of tokens sampled from a glyph trigram model that occur in the training half. Dotted lines separate the Voynich text from the controls.

3.3 A local glyph process suffices; the published mechanisms do not

Table 1. The profile battery on the Voynich text, Latin controls and candidate generators, all in the line and token shape of the Voynich paragraph text. Boundary MI: excess MI between the last glyph of a token and the first glyph of the next; token MI: between whole adjacent tokens (both against within-line permutation); line-break MI: (last token, first token) across line breaks; space gain: information about token boundaries from one glyph on each side; attested: share of held-out real tokens and of tokens sampled from a glyph trigram model found in the training half; residual: lzma bits per glyph minus that of an order-3 section-drift twin (negative: redundancy beyond the automaton). Full table and variants in Tables S7, S8 and S9.

Text or generator	boundary MI	token MI	line-break MI	space gain	hapax	attested real	attested sampled	residual	I(token; page)
Voynich (ZL3b)	0.146	0.078	0.005	73.7%	0.71	0.79	0.75	-0.039	0.232
glyph automaton, order 3, section drift	0.167	0.050	0.005	73.8%	0.71	0.79	0.76	$+0.033$	0.134
glyph automaton, order 4, page drift*	0.158	0.076	0.004	73.8%	0.63	0.81	0.75	-0.052	0.215
Latin, running text	0.034	0.245	0.076	38.9%	0.59	0.80	0.20	-0.397	0.062
Latin, one sentence per line	0.035	0.646	0.159	38.7%	0.35	0.86	0.22	-0.993	0.050
table and grille (after Rugg)	0.006	0.613	0.041	62.8%	0.41	0.73	0.55	-0.766	0.049
copy-and-modify (after Timm and Schinner)	0.010	0.022	0.042	46.7%	0.62	0.57	0.23	-1.030	0.912
verbose cipher, fixed homophones	0.409	1.237	0.136	47.4%	0.30	0.93	0.64	-0.950	0.015

ciphertext-feedback substitution	0.649	0.160	0.006	36.5%	0.78	0.39	0.27	−0.310	0.003
Naibbe cipher (Latin)	0.003	0.055	0.002	77.4%	0.49	0.85	0.72	−0.065	0.007
Naibbe, state-dependent tables, $w = 0.9$	0.085	0.181	−0.001	76.7%	0.48	0.86	0.64	−0.148	0.004
Naibbe, state-dependent tables, $w = 1.0$	0.255	0.342	0.001	75.4%	0.49	0.87	0.66	−0.222	0.002

* fitted in-sample (the page-level counts include the page itself); the out-of-sample evidence is in Table S11.

Table 1 applies the battery to candidate generators. All of them produce text in the exact line and token shape of the Voynich paragraph text. A glyph automaton of order 3 is fitted per Currier language and section, treats the space as a symbol, and is sampled line by line. It reproduces the boundary coupling (0.167 against 0.146), the line-break reset, the boundary predictability, the hapax share (0.71 against 0.71), the lexicon test and the residual. Its token MI (0.050 against 0.078) and its page locality (0.134 against 0.232) fall short. Letting its parameters drift by page closes both gaps (0.076, 0.215), but that variant is fitted in-sample.

Out of sample, an automaton fitted per section on the odd-numbered pages matches the held-out even pages in boundary coupling (0.150 against 0.150), hapax share and the lexicon test, with residuals of −0.030 (order 3), +0.020 (order 4) and +0.010 bit per glyph (halves swapped). It misses only the token MI (0.007 against 0.034) and the page locality (0.168 against 0.289), which is the page-level variation a section-level model cannot contain (Table S11). The profile holds on the Takahashi transliteration (boundary MI 0.165), with uncertain spaces treated as spaces (0.182), and in each of the three main hands (0.079, 0.195, 0.185; Tables S10–S12).

The other mechanisms fail in characteristic ways.

- **Table and grille; copy-and-modify.** Both choose the parts of neighbouring words independently, so they have almost no boundary coupling (0.006 and 0.010 bit). Table reuse and copying create long-range repetition that the compressor finds (residuals −0.766 and −1.030).
- **Verbose cipher with fixed homophones.** It leaks the collocations of its plaintext (token MI 1.237) and has few hapaxes (0.30).
- **Ciphertext-feedback substitution**, the class that can in principle imitate a Markov chain, overshoots the boundary coupling (0.649), makes spaces unpredictable (36.5%) and keeps the redundancy of its plaintext (residual −0.310).
- **The Naibbe cipher** is the strongest competitor. It matches the entropy profile, the space predictability (77.4%) and, unlike every other plaintext mechanism, the residual (−0.065): the entropy of its card draws hides the plaintext's redundancy from the compressor. It fails the boundary coupling by a factor of about fifty (0.003), the hapax share (0.49, from an inventory closed at 5,978 types) and the page locality (0.007). Our reimplementations reproduces the author's own implementation. Enciphering the author's tokenised plaintext, 100.0% of his 34,764 published ciphertext tokens are valid outputs of the tables, the table shares lie within 0.014 of the deck proportions, and 7 of 8 statistics agree with his output; only the hapax share differs slightly (0.505 against 0.518). On the author's own ciphertext, the boundary MI is 0.0010 and the page MI 0.000 (Tables S36 and S37).
- **State-dependent Naibbe.** With probability w the next table is chosen from the last glyph of the previous token. This creates boundary coupling but costs key entropy. At $w = 1.0$ the coupling (0.255) exceeds the manuscript's, while the token MI rises to 0.342 and the residual to −0.222. At $w = 0.9$ the coupling is still too small (0.085) and the token MI

(0.181) already too large. For a cipher of this kind, boundary coupling and hidden plaintext redundancy compete. For a glyph automaton without plaintext they do not.

The table-and-grille and copy-and-modify generators are our reconstructions of the published descriptions; unlike the Naibbe cipher, they could not be checked against their authors' outputs. Their failures follow from the constructions (independent choice of neighbouring boundary glyphs; long-range reuse), not from parameter settings.

3.4 Drift, scribal hand and the illustrations

Drift and hand. The glyph-bigram distributions of two pages of the same section and Currier language diverge more the farther apart the pages are in the binding. In Currier A the Jensen–Shannon divergence rises from 0.124 at one or two pages to 0.161 beyond forty, while a twin with one model per section stays flat (0.093 to 0.094; Table S13). Sections occupy contiguous folio ranges, so section and drift are confounded. In the one section written by several hands (Herbal-B, 32 pages), pages at most six folios apart differ by 0.138 when written by the same hand and by 0.182 when written by different hands (Table S14). At fixed section, language and distance, the hand effect exceeds the section effect, which makes scribal practice a plausible source of the drift. That hands and Currier languages coincide is known (Currier 1976; Davis 2020); the controlled comparison is new.

Text and illustration. If the text of a herbal page described its drawing, pages with similar drawings should carry similar text. We compared text distance and colour-feature distance between pages with a Mantel test (Mantel 1967), excluding pairs closer than three folios. For the 85 Herbal-A pages of hand 1 the correlation is $r = -0.055$ ($p = 0.79$) for token frequencies and $r = -0.070$ ($p = 0.87$) for glyph bigrams. The same text distance does detect the folio drift ($r = +0.115$, $p = 0.001$; Table S15). A positive control adds to each page k pseudo-tokens that are deterministic functions of its features; the test detects them from $k = 10$ ($r = +0.195$, $p = 0.001$; Table S16), about 11% of a page's tokens. An unstratified correlation in Herbal-B ($p = 0.046$) is carried by the hand and disappears when the permutation is stratified within hand ($p = 0.076$). The test therefore excludes descriptions in which about a tenth or more of a page's tokens are determined by the measured colours. It says nothing about content that is not visible in the colours, such as names or uses.

Labels. Label tokens recur on other pages of their section less often than paragraph tokens of the same pages (zodiac 0.276 against 0.363; pharmaceutical 0.090 against 0.364; Table S19). Of 271 zodiac label types, 36 occur on two or more of the twelve sign pages and none on more than 4, so they are not numerals of days or degrees. Labels behave as unique identifiers.

3.5 What the profile establishes

All tested structure is compatible with a local, low-order glyph process whose parameters drift by section, page and hand. Neither a lexicon nor information about the colours of the drawings is measurable beyond it. This is a statement of sufficiency: the automaton is a positive null model, not a historical procedure, and a cipher whose free choices are driven by a plaintext could leave the same footprint (Section 8). The profile leaves open which mechanisms the text actually excludes. The pre-specified program that follows addresses this and finds properties the automaton lacks: short-range recency, within-page drift, line-final forms that depend on whether the line ends at the margin, and very low recurrence of locally deviant forms.

4. Framework and protocol

4.1 Hypotheses

Table 2. Hypotheses and their operational definitions as pre-specified. The two axes are independent.

Axis	Hypothesis	Operational definition	Signature predicted in advance
Generation	P, fixed procedure	explicit rules plus an external randomiser or schedule; no free choices; parameters change only at discrete points	hard zeros; stationarity within a setting; possible device periodicity; slip-like deviations
	I, skilled improvisation	internalised, soft habits of a writer producing pseudo-text online	priming; smooth drift; innovations that recur; sensitivity to layout
	C, plaintext-driven procedure	explicit rules whose choices are driven by a plaintext (natural language as identity mapping)	content-driven recurrence and topic structure; no fixed prediction for randomised encodings
Transmission	D	composed directly on the page	line-aware forms tied to the page's own lines
	K-reflow	copied from a draft with different line breaks	no line-final forms; breaks as transparent as word boundaries
	K-same	copied from a draft with the same line breaks	not separable from D by the transliteration (stated in advance)

The copy-and-modify account is classed as a formalised variant of improvisation, because its source and modification choices are made by the writer. Natural language is the degenerate plaintext-driven case with an identity mapping. Tests based on the transliteration alone cannot separate direct composition from copying with identical line breaks, and none of our tests claims to.

4.2 Claims policy

Three rules were fixed before any test:

- Statistical sufficiency of a model is not historical identification of a mechanism.
- No result is reported as showing that the text is meaningless or contains no plaintext. The strongest admissible statement about the plaintext hypothesis is that a named variant of it makes a prediction that fails.
- A concrete generator is reconstructed only if the verdict rule favours a fixed procedure.

4.3 Pre-specification and stages

We call an analysis *pre-specified* when its hypotheses, statistics, predictions, falsifiers and calibration requirements were written down and committed to the project's git repository before its Voynich outcome was computed. This is prospective specification with version control, not registration in an external registry such as OSF. The commits are listed in Table S28. The repository was private while the work was done and was published when it was complete, so the timestamps are the author's record and cannot be verified independently. Before publication the author e-mail address was removed from all commits; this changed the commit identities but not their contents or dates (mechanism/COMMIT_MAP.md). The battery ran in two stages: controls only (Stage A, committed), then the Voynich text (Stage B). Four further analyses answer a review of an earlier version (R1–R4, mechanism/REVISION_PLAN.md): R2–R4 were pre-specified before computation, and R1 is a post hoc check.

Deviations were logged with the reason and the time relative to the Voynich run (Table S27). All were made before the affected Voynich statistic was computed. The main ones:

- **T1 null model.** The pre-specified corpus-wide re-pairing null for line-break coupling was confounded by writer heterogeneity. It was replaced by a page-conditional estimator that we validated on a known-truth simulation.
- **E1 expectation.** Position-stratified expectations were added after a control without settings showed a spurious step.
- **E5 intervals.** A page bootstrap was replaced by a quire jackknife, because duplicated

pages bias same-page pair counts.

Limits of blindness. The author knew several Voynich properties from the exploratory profile (Section 3); they are declared in the battery's analysis plan. While diagnosing the T1 estimator, its terms were computed once on the Voynich text; that test's outcome was already known and is reported as "prior". After E3 the Voynich values of both halves had been computed, so later generator comparisons were blind only in their fitted parameters, not in the values they were compared with.

4.4 Calibration

A test counts toward a verdict only if it separates controls of the classes it is meant to separate (Table 3). Two pre-specified predictions were contradicted by their own controls and were narrowed rather than applied:

- All four natural languages drift within a page. The within-page drift test (T6b) therefore separates only stationary procedures from the rest.
- The Naibbe cipher reuses its rare forms as seldom as the procedures do. The deviant-token test (T7b) therefore excludes only word-preserving plaintext systems.

Table 3. Calibration requirements on controls with known mechanism. A falsifier is applied only when its calibration holds.

Test	Calibration requirement	Outcome
T1 break coupling	greedily wrapped Latin $R \geq 0.5$	$R = 2.44$ (SE 2.75): met, very noisy
T2a final form	no effect in wrapped Latin	excess 0.0000, $p = 0.46$: met
T2b line fill	wrapped Latin T within 0.9–1.1	$T = 0.994$ (0.948, 1.037): met
T4 periodicity	power against a periodic device	mixing weight 0.1: 2/10, 0/10, 0/10; 0.2: 10/10, 5/10, 10/10; 0.4: 10/10, 10/10, 10/10 (lag, column, line-cycle tests)
T5 drift shape	smooth control $S > 1$, step control $S < 1$	1.66 and 0.28: met
T6a recency	stationary procedures near 0	highest 99th percentile 0.116: met
E1 boundary steps	settings controls step, recency control does not	met after position stratification (deviation, before the Voynich run)
E4 drift	≥ 3 languages with lexical drift; glyph-drift control sublexical; vocabulary-drift control lexical only	met
E5 burstiness	≥ 3 languages with positive gradient; Naibbe closer to 1	met (quire jackknife; deviation, before the Voynich run)

5. The pre-specified battery

5.1 Tests

- **T1, line-break transparency.** The ratio of the excess mutual information between the last unit of a line and the first unit of the next to that between adjacent words inside lines. It is computed per page with a re-pairing null and averaged.
- **T2a, final form.** The Jensen–Shannon divergence between the final-unit distribution of line-final words and that of line-internal words of the same length. It is computed separately for margin-bound and paragraph-final lines, against a permutation null.
- **T2b, line fill.** Slack at the line end relative to greedy packing of the same page's words.

- **T2c, dittography.** Repetition of words across line breaks.
- **T3a, zero replication.** In matched 250-token chunks, how often unit bigrams that are absent in one half although expected (independence model) appear in the other half.
- **T3b, slot rigidity.** The share of ordered unit pairs inside words that is consistent with the best single linear order of units.
- **T3c, positional entropy.** Entropy of the unit at each position from the start and the end of the word.
- **T4, periodicity.** Lag, column-alignment and line-cycle profiles, with power measured on a periodic device.
- **T5, drift shape.** Step against smooth models of page-to-page distances within a hand.
- **T6a, recency.** Exact and near (edit distance 1) repeats at short against long token distance, relative to within-page permutation.
- **T6b, within-page drift.** The slope of line similarity over line distance.
- **T7a, annotated corrections.**
- **T7b, deviant tokens.** Double-coded tokens whose unit trigrams occur in no other quire, and whether they recur.

5.2 Results

Table 4. The pre-specified battery on the Voynich text (Stage B) with the controls that calibrate each reading. Intervals are 95%.

Test	Voynich (ZL3b)	Takahashi	Key controls	Reading
T1 line-break ratio	0.22 (SE 0.25)	0.18 (0.18)	gibberish 1.10 (0.32); automaton – 0.14; copy-and-modify 1.10	strongly reduced coupling; K-reflow falsifier not triggered
T2a final form, margin	0.085, p = 0.005	0.094	gibberish 0.0033; wrapped Latin 0.0000	K-reflow falsified
T2a final form, paragraph end	0.039, p = 0.005	–	gibberish 0.034	weaker than at the margin
T2b line-fill slack	2.09 (1.88, 2.32)	–	gibberish 3.90 (3.11, 4.81)	more regular fill than naive online writers
T2c dittography	0.80, p = 0.77	–	copy-and-modify 1.38, p = 0.01	no copying signature
T3a zero replication (250 tokens)	0.040 (0.038, 0.043)	0.039	gibberish 0.093; languages 0.075–0.104; generators 0.027–0.052	harder at short range; naive improvisation falsified
T3b slot consistency	0.868 (0.863, 0.873)	0.885	gibberish 0.671; languages 0.66–0.70	rigid slot order
T3c final-position entropy	0.63	–	Naibbe 0.65; automaton 0.67; languages 0.78–0.91; gibberish 0.98	collapse (Kinnison 2026) reproduced by cipher and automaton
T4 periodicity	no peak	no peak	device detected from weight 0.2	device variants ≥ 0.2 excluded
T5 drift shape S	0.76	0.81	smooth 1.66; step 0.28; section automaton 0.62	step-like (hands 2 and 3)
T6a recency (near)	0.153 (0.112, 0.198)	0.143	stationary 99th pct ≤ 0.116 ; copy-and-modify 0.34; English 0.10	stationary procedures falsified

T6b within-page drift	−0.0083 (−0.0096, −0.0073)	−0.0082	automaton −0.0004; languages −0.012 to −0.003	drift of natural-language strength
T7b deviant recurrence	1.1% (slip share 33%)	1.8%	languages 5.5%–16.1%; copy-and-modify 32%; automata 7.7%–9.2%; Naibbe 2.6%–3.8%	rare forms not reused

Takahashi values are post hoc replications with identical code.

Table 4 summarises the battery. Four results carry most of the weight.

Rigidity and hardness. Slot order within Voynich tokens is far more rigid than in any natural language or in the human gibberish (Figure 2). Each gibberish writer's document is scored separately, because every writer invented a different "language"; documents with at least 150 tokens (36 of 38) are the units, and the Voynich text is scored in 250-token stretches within a hand. The mean slot-order consistency is 0.868 against 0.671 for the gibberish writers and 0.66–0.70 for languages. The contrast holds writer by writer. Compared with contiguous Voynich stretches of the same length within one hand, all 36 of 36 writers lie below the 5th percentile of the Voynich distribution; the writers range from 0.57 to 0.78 (R1, post hoc; Table S35). Its share of words fully monotone in the best order is 0.55 against 0.24 and 0.12–0.35.

Within 250-token stretches, zeros of the unit-transition table replicate far more consistently: the zero-replication ratio is 0.040 against 0.093 and 0.075–0.104. This hardness contrast is scale-dependent. At full corpus size all corpora show hard zeros (Voynich 0.0043, languages 0.0017–0.0042). What distinguishes the Voynich text is that its constraints are expressed within any short stretch, not that it has more of them.

Both contrasts survive raw EVA characters without multigraph merging (Voynich 0.034 and 0.856; gibberish 0.098 and 0.679) and hold in each main hand. The pre-specified falsifier of naive improvisation is triggered. Skilled, long-practised improvisation cannot be falsified by this test, because no data on such writers exist.

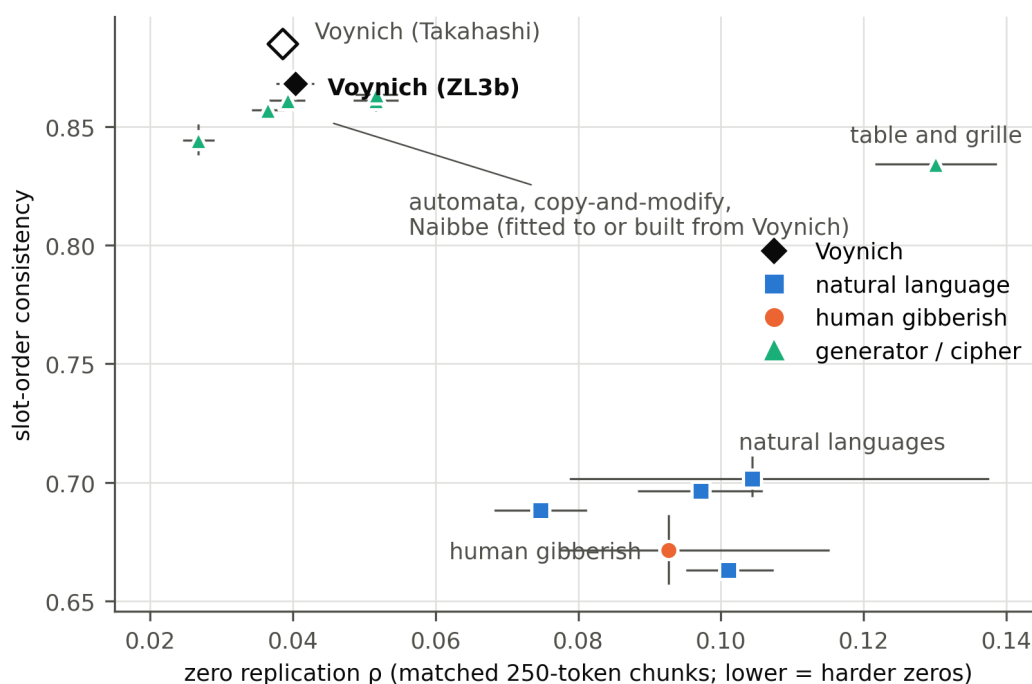


Figure 2. Hardness of the unit-transition zeros (T3a, lower is harder) against slot-order rigidity (T3b), with 95% intervals. Generators and ciphers derived from Voynich statistics inherit its structure. Human gibberish sits with the natural languages.

Line endings. Line-final tokens have their own final-glyph distribution, beyond what their length explains (T2a: 0.085, $p = 0.005$, $n = 3,373$; Takahashi 0.094). Greedily re-lined Latin shows none (0.0000), so copying a running text with new line breaks is falsified.

The effect is weaker at paragraph ends (0.039). The difference, 0.046 (0.026, 0.062), was checked post hoc by a page bootstrap. On the text-only pages of the stars section, where every non-final line ends at the text block's margin, the contrast is sharper: 0.156 against 0.033. The line-final forms are therefore associated with lines that end at the margin, not with line ends as such (Figure 3). The measurement shows the association; that the writer chose the last word to fit the space is an interpretation (Section 9.4).

The online gibberish writers show almost no margin effect (0.0033). Line fill is more regular than theirs (T2b 2.09 (1.88, 2.32) against 3.90). Glyph count is, however, a noisier proxy for the width of handwritten Latin letters than for Voynich glyphs, so we read T2b only as "more regular fill".

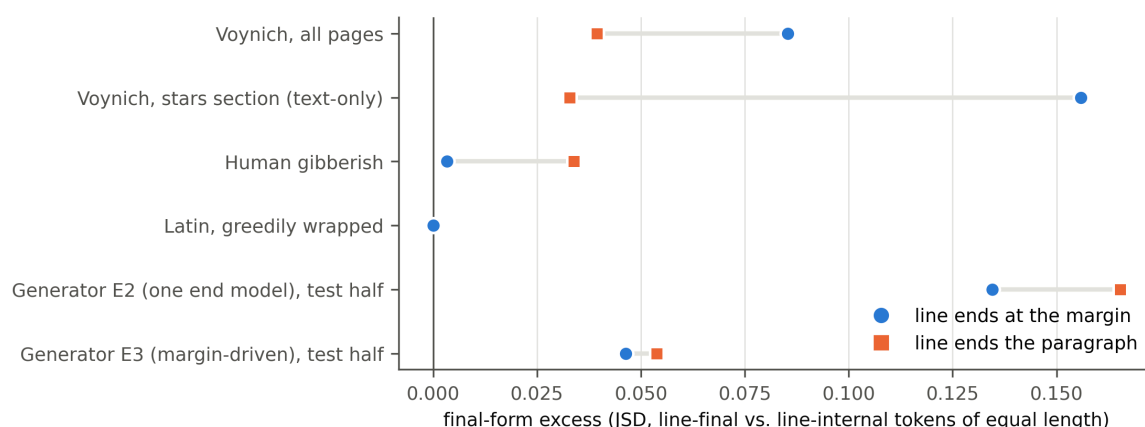


Figure 3. Final-form excess for lines that end at the margin and lines that end a paragraph. The E2 generator has a single end model and makes every line end special. E3 separates the two kinds of line end. Generator values are on the held-out half.

Recency. Near repeats are more frequent at short token distance than at long distance within the page (T6a: 0.153 (0.112, 0.198); exact repeats 0.217 (0.131, 0.303)). This is above the highest 99th percentile of six stationary procedures (0.116), so stationary procedures are falsified. It is about half the strength of the copy-and-modify model (0.34) and similar to the strongest language control (English, 0.10).

The human gibberish does not show significant recency (0.071 (−0.099, 0.235)). Its documents average about 265 tokens, so the improvisation prediction is calibrated mainly by the formal copy-and-modify model.

Deviant forms. Double-coded tokens containing a unit trigram found in no other quire occur at 14.7 per thousand and recur in 1.1% of cases (1.7% without double coding). The corresponding values are 5.5%–16.1% in the languages, 32% for copy-and-modify and 7.7%–9.2% for the automata. One third of them lie at edit distance 1 from a common form of the same quire, as slips would.

The result holds with page instead of quire as the block (0.5% against 2.1%–5.1% in the languages) and in each hand (1.1%–2.5%). The Naibbe cipher is equally low (2.6%–3.8%; page level 0.4%). This property therefore excludes the tested natural-language controls and systems that preserve lexical identity at the token boundaries, not homophonic ciphers.

Other tests.

- **Periodicity:** no peak in any profile. The device control (period 5, cycle 4) is detected in 10 of 10 seeds from mixing weight 0.2 (Table 3); periodicity of other designs is not tested.

- **Drift shape:** step-like ($S = 0.76$; calibration 1.66 smooth, 0.28 step). Per hand it is 1.07, 0.59 and 0.68, so hand 1 alone is smooth-like. In hand 2 the largest jumps fall at quire changes (mean excess 0.080) rather than section-only changes (0.028, like no change at 0.030). This decomposition is post hoc and rests on few boundaries.
- **Within-page drift:** as strong as in the languages (-0.0083 against -0.012 to -0.003).
- **Line breaks:** the break ratio is 0.22 (SE 0.25); the gibberish writers are transparent across breaks (1.10).
- **Dittography:** none (T2c).
- **Annotated corrections (T7a):** the 24 flagged in the transliteration are all glyph-level. The pre-specified position test turned out to measure an annotation convention (comments follow the corrected word), so it is uninformative.

5.3 Verdict of the battery

Under the pre-specified scoring the fixed procedure collected the most support. It was vetoed by the recency test, because the text shows local dynamics no stationary procedure produces. Improvisation was supported by recency and contradicted by the drift shape and the deviant forms. The tested plaintext variants each failed a test. The verdict was therefore a bounded, unresolved competition among three classes:

- a procedure with state below the page level;
- a practised improviser with an internalised hard grammar;
- a plaintext-driven procedure with rigid tables and recency-dependent choices.

The experiments of Section 6 were designed to separate them.

6. Experiments that followed the battery

6.1 E1: settings or continuous recency?

Local dynamics can arise from settings that stay fixed within a line or paragraph and change at its boundary. They can also arise from continuous recency, in which a form becomes more likely after a similar form a few words earlier. The first predicts steps in pair similarity at boundaries and no decay inside a segment; the second predicts the reverse.

E1 compares pairs of line-internal words at matched distance within a line, across a line break, and across a paragraph break. Expectations are stratified by position class. Settings controls were tuned to the known Voynich recency index (line settings $\sigma = 0.5$, paragraph settings $\sigma = 0.6$, recency ρ , τ , $e = \rho = 0.05$, $\tau = 8$, $e = 0.1$).

Those controls produce clear steps (line -0.086 graded, paragraph -0.25). The Voynich text shows none (Table 5, Figure 4b). Its graded line step is $+0.054$ (0.026, 0.080); line settings are excluded because the lower bound lies above the pre-specified threshold of -0.039 . Its paragraph step is -0.039 (-0.102 , 0.019); paragraph settings are excluded, marginally, because the lower bound lies above the threshold of -0.116 . Decay inside lines exceeds every control (0.069 (0.037, 0.101) against -0.05 to 0.01). Both transliterations agree.

Settings that switch per line or per paragraph, at the magnitude needed for the observed recency, are excluded. The dynamics are continuous: memory spans line breaks, while the glyph-level boundary coupling resets at each line. A small paragraph effect on top of recency, such as a pause or a change of topic, is not excluded.

Table 5. E1: steps in pair similarity at line and paragraph boundaries at matched token distance, with position-stratified expectations. Control values are means over 5 seeds; intervals are 95% page-bootstrap intervals.

Statistic	Voynich	Takahashi	Settings per line	Settings per paragraph	Recency	Automaton
line step, graded	+0.054 (0.026, 0.080)	+0.066	−0.086	−0.016	−0.023	0.015
line step, binary	+0.010 (−0.074, 0.085)	−0.019	−0.222	−0.034	−0.009	−0.035
paragraph step	−0.039 (−0.102, 0.019)	−0.034	−0.007	−0.247	0.004	−0.021
within-line decay, graded	0.069 (0.037, 0.101)	0.076	−0.026	−0.029	−0.032	−0.010
within-paragraph decay	0.065 (0.016, 0.110)	0.074	−0.018	0.016	0.072	−0.019

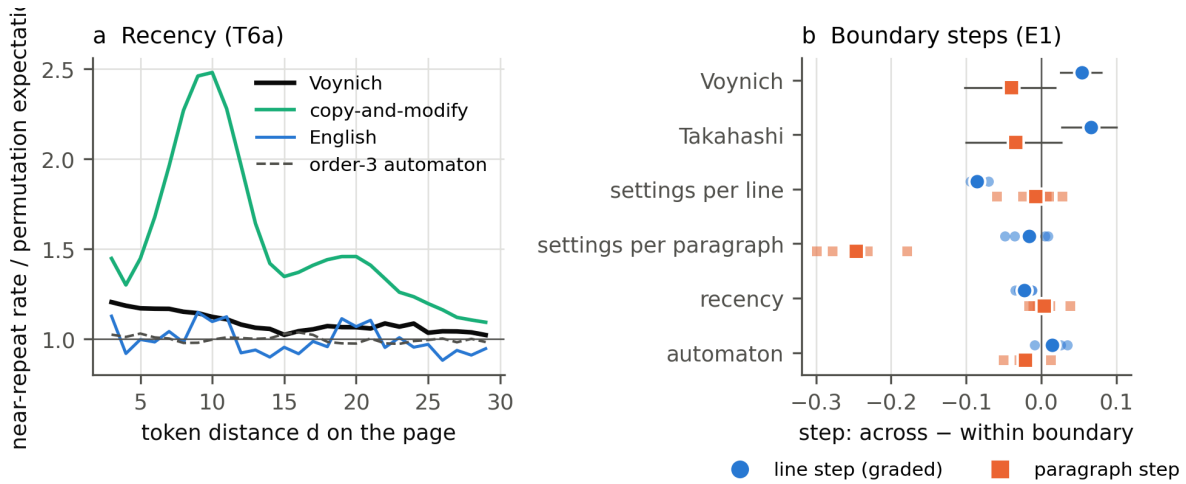


Figure 4. (a) Near-repeat rate relative to within-page permutation by token distance (three-point smoothing). (b) Steps in pair similarity at line and paragraph boundaries. Settings controls step and the Voynich text does not. Small dots are single control seeds.

6.2 E2, E3 and E6: explicit generators on held-out pages

E2 built a generator of the surviving class:

- a hard-pruned order-3 character automaton per group, restarted at each line;
- a quire-level session weight and a page-level tilt of glyph preferences;
- continuous recency;
- slips.

It was fitted on the odd bifolios against five targets and tested on the even bifolios against 40 statistics. These are the profile statistics of Section 3, T1–T7 and E1. A statistic passes if the generator's mean lies within $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$ of the Voynich value, with SE from a quire jackknife and SD over three seeds.

E3 added the three components whose absence E2's failures pointed to:

- separate line-end models for margin-bound and paragraph-final lines, with tight length control and a final word chosen so the line can end there;
- a random walk of glyph preferences from line to line;
- slips that are never reused, with memory keeping the intended form, plus a choice temperature.

E3 was cross-fitted: its primary direction was fitted on the even half and tested on the odd half, whose Voynich values for 35 of the 40 statistics had not been computed. E6 tested a two-route

alternative motivated by E4 and E5: whole familiar forms retrieved from a drifting repertoire, plus one-off coinages from the grammar.

Table 6. Explicit generators of the surviving class, fitted on one half of the manuscript (bifolio parity) and tested on the other against 40 statistics.

Exp.	Generator	Held-out pass (primary)	Secondary	Failing statistics (primary)
E2	hard grammar + recency + session state	32/40	–	hapax share; type-token ratio; space predictability; lexicon test, real words; T2a final form, paragraph end; T2b line-fill slack; T6b within-page drift; T7b deviant recurrence
E3	+ margin-driven endings, slips, drift, temperature	33/40 (28/31 unfitted)	28/40	word-pair MI; lexicon test, synthetic words; T2b line-fill slack; T3a zero replication; T6b within-page drift; T7b deviant rate; T7b deviant recurrence
E6	two routes: drifting repertoire + one-off coinage	22/40	29/40	hapax share; type-token ratio; mean word length; space predictability; lexicon test, real words; lexicon test, synthetic words; conditional entropy h2; conditional entropy h3; conditional entropy h4; lzma bits per character; ...

E2 was fitted on odd and tested on even bifolios. E3 and E6 were cross-fitted, and their primary direction (fit on even, test on odd) used Voynich values not computed before.

	E2 full	E2 plain automaton	E3 full primary	E3 full secondary	E3 w/o margin rule	E3 w/o slips etc.	E6 two-route	E6 w/o boundary
hapax share †	×				×	×	×	×
type-token ratio †	×					×	×	×
Zipf slope								
mean word length							×	×
boundary MI								×
word-pair MI			×	×				×
distance-2 word MI		×		×				
line-break word MI								×
line-break boundary MI								
boundary MI, equal n								
space predictability	×			×		×	×	×
lexicon test, real words	×						×	×
lexicon test, synthetic words			×				×	×
conditional entropy h2				×		×	×	×
conditional entropy h3				×		×	×	×
conditional entropy h4				×		×	×	×
lzma bits per character							×	
page MI †		×						
T1 line-break ratio								×
T2a final form, margin					×			
T2a final form, paragraph end	×	×			×			
T2b line-fill slack	×	×	×	×	×	×	×	×
T2c dittography								
T3a zero replication †		×	×	×	×	×		
T3b slot consistency							×	×
T3b monotone share							×	×
T3c final-position entropy								
T3c initial-position entropy								
T4 periodicity								
T5 drift shape								
T6a recency, near repeats †		×		×		×	×	×
T6a recency, exact repeats †						×	×	×
T6b within-page drift †	×	×	×	×	×	×	×	×
T7b deviant rate †			×	×				
T7b deviant recurrence †	×	×	×	×	×	×	×	×
T7b slip share					×	×	×	×
E1 line step								
E1 paragraph step								
E1 within-line decay								
E1 within-paragraph decay								
	32/40	32/40	33/40	28/40	32/40	27/40	22/40	19/40

Figure 5. Held-out pass (blue) and failure (×) for each statistic and generator. † marks statistics used as fitting

targets on the training half. Totals are at the bottom.

What the generators established (Figure 5). The best generator (E3) passes 33 of 40 statistics (28 of the 31 that were not fitting targets). In the secondary direction it passes 28. The ablations identify necessary components:

- **Recency:** without it, conditional entropies and the recency indices fail.
- **Hard pruning:** without it, zero replication and distance-2 MI fail.
- **Quire-level state:** without it, zero replication, distance-2 MI and near recency fail.
- **Margin-conditioned end model:** without it, the final-form statistics fail, and line-fill slack rises from 2.63 to 8.07 (Voynich 2.27).
- **Glyph-level boundary dependency:** without it, a word-retrieving generator loses the boundary coupling entirely (edge MI 0.0005 against 0.139).

Two properties defeated every generator:

- **Very low recurrence of deviant forms.** Deviant forms recur in 0.9% of cases in the Voynich half and in 4.7%–18.7% across all generators.
- **Within-page drift.** Voynich -0.0084 against -0.0021 to -0.0002 .

A glyph-level random walk of preferences, strong enough to matter, degrades other statistics before it reaches the observed drift. Slips raise lexical diversity to the observed level but soften the zeros (T3a 0.057 against 0.042). The text is diverse while staying hard, which suggests that its novel forms are grammatical constructions rather than errors.

The two-route generator failed clearly (22/40). Retrieval from a drifting repertoire collapses lexical diversity (type-token ratio 0.13 against 0.29). This prediction, derived from E4 and E5, is not supported.

6.3 E4 and E5: where the drift and the page effects live

E4 decomposes the within-page drift into a lexical part, meaning which word types two lines share, and a sublexical part, meaning how the remaining words are built. For the sublexical part, words within edit distance 2 of each other are removed from the comparison. Slopes are taken over line distances 2–8, which excludes token-level recency.

The Voynich drift is lexical in type ($L = -0.0142$ ($-0.0275, -0.0022$)) with a small sublexical part ($S = -0.0052$ ($-0.0085, -0.0023$)). That is the profile of plain Latin (-0.0028) and German (-0.0043). The pre-specified bar for "sublexical beyond content drift" was set by Dante's terza rima (-0.0153), and it was not met, so E4 does not separate content drift from vocabulary or habit drift.

A comparison of pre-specified statistics adds one point. Both ciphers of Latin remove the sublexical drift that their plaintext has (deterministic verbose cipher $+0.0002$ ($-0.0020, 0.0022$); Naibbe -0.0015). The Voynich interval lies below the deterministic cipher's, so letter-level ciphers of the tested kinds are poor sources of the observed drift.

E5 asks whether the page clustering of word types is content-like. Content drift makes rare content words the burstiest (Church and Gale 1995): a name or term clusters on the pages about its subject. E5 compares observed same-page pairs with their expectation under permutation within hand $\times$ section, by frequency band. The gradient G contrasts rare (2–4) with frequent (20–99) types.

The languages give $G = +0.74$ to $+1.08$, the deterministic cipher $+0.66$ and Naibbe $+0.07$. The Voynich value is $+0.245$ ($-0.099, 0.588$) (Takahashi $+0.189$), which is undetermined by the pre-specified rule; E5 is therefore used only descriptively. Its point estimate lies below the lower bound of every content control. It clusters by page in every band, including forms occurring at least 100 times (1.36 (1.17, 1.56) against 1.02–1.22 in the languages).

So the page effect sits in the familiar forms, which natural-language content drift leaves nearly uniform. A glyph-level generator with page tilts reproduces the clustering of frequent forms but overshoots on rare ones ($B = 3.68$ for rare forms, against 2.02). Part of what earlier work read as keyword clustering (Montemurro and Zanette 2013) may be this production-level page effect. Our test measures clustering by page within hand and section, not their section-level measure, so the two are not directly comparable.

Table 7. E4 and E5: where the within-page drift and the page clustering live. Intervals in Tables S33 and S34.

Corpus	Lexical drift L (E4)	Sublexical drift S (E4)	Burstiness gradient G (E5)	Clustering of forms ≥ 100 (E5)
Voynich	-0.0142	-0.0052	+0.24	1.36
Latin	-0.0115	-0.0028	+0.74	1.07
Italian	-0.0147	-0.0121	+0.78	1.02
German	-0.0252	-0.0043	+1.06	1.13
English	-0.0128	+0.0003	+1.08	1.22
verbose cipher (Latin)	-0.0092	+0.0002	+0.66	1.03
Naibbe (Latin)	+0.0076	-0.0015	+0.07	1.00
glyph-habit drift	-0.0517	-0.0148	-	-
vocabulary drift	-0.0398	+0.0007	-0.19	2.12
repertoire + coinage	-	-	-0.60	2.40
glyph generator (E3)	-	-	+0.97	1.31
order-3 automaton	-0.0042	-0.0007	+0.03	1.03

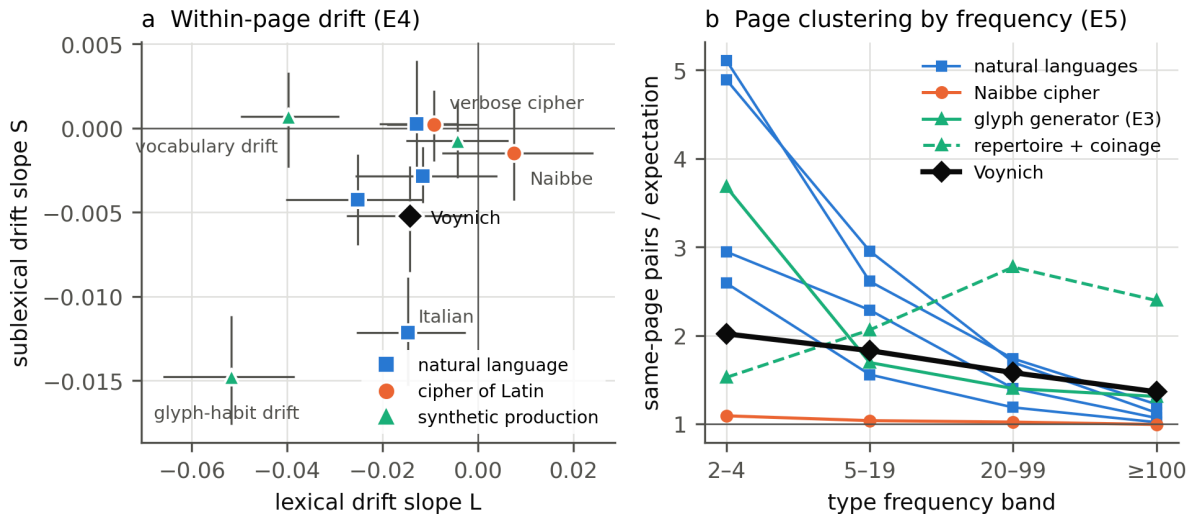


Figure 6. (a) Lexical against sublexical drift slopes (95% intervals). (b) Page clustering by frequency band: observed same-page pairs relative to permutation within hand $\times$ section.

7. The mechanism class

We define the mechanism class as the set of stochastic processes that, on the manuscript's layout, produce token sequences satisfying the nine properties in Table 8, as measured by the stated statistics. The class is defined by necessary conditions.

It is not yet shown to have a member that reproduces all 40 statistics: the best explicit member

reaches 33, and the two open specifics are the very low recurrence of deviant forms together with the within-page drift. The strength labels in Table 8 reflect the red-team checks of Table 12: robustness to transliteration, segmentation, deviance definition, hand and sample size.

Table 8. The nine properties that define the mechanism class, stated as measured, with their evidence and an explicit strength label. Causal readings of these properties are discussed in Section 9.

#	Property	Evidence	Strength
C1	Rigid slot order within words; transition zeros consistent within short stretches	T3b 0.868 vs gibberish 0.671, languages 0.66–0.70; T3a (250 tokens) 0.040 vs 0.093, 0.075–0.104; pruning necessary (E2)	strong (T3b); moderate (T3a: not at full size)
C2	Cross-token boundary dependency within lines (last glyph of a token, first glyph of the next), strongly reduced across line breaks	boundary MI 0.139; T1 ratio about 0.03 (pooled) to 0.22 (page-conditional); boundary rule necessary (E6: 0.0005 without)	strong within lines; moderate at breaks
C3	Margin-associated line-final forms, weaker at paragraph ends	T2a 0.085 vs 0.039, difference 0.046 (0.026, 0.062); stars section 0.156 vs 0.033; necessary in E3	strong
C4	Short-range recency, continuous across line and paragraph boundaries	T6a 0.153 > 0.116; E1 no steps; within-line decay 0.069; necessary in E2	strong
C5	Page- and quire-level variation in the weights of shared forms	T5 $S = 0.76$; E5 very frequent forms clustered 1.36 vs languages 1.02–1.22 (descriptive); quire-level state necessary in the generator (E2)	moderate (hand 1 smooth)
C6	Within-page drift of natural-language strength, mainly lexical	T6b -0.0083 ; E4 $L = -0.0142$, $S = -0.0052$	strong (existence); undetermined (source)
C7	Very low recurrence of locally deviant forms	T7b 1.1% vs languages 5.5%–16.1%, copy-and-modify 32%, generators 4.7%–18.7%; page-level variant 0.5%	strong; shared with homophonic ciphers
C8	No periodicity of the tested kind	T4, power from mixing weight 0.2 (period 5, cycle 4)	moderate (tested design only)
C9	No detected copying signature; copying from a re-lined draft excluded	T2c 0.80; no word-level corrections among 24; T2a	moderate (T7a thin)

Table 9 separates three kinds of outcome.

- **Exclusions** are variants with a named falsifier triggered under calibration.
- **Unresolved alternatives** are variants the text cannot decide between. For two of them this is shown in principle; for the third, the transliteration lacks the information.
- **Open specifics** are properties no tested generator reproduces jointly.

Table 9. What is excluded, and what remains open. "Excluded" means that a pre-specified falsifier was triggered under calibration; it applies to the variant as defined and tested. "Unresolved" entries are shown to be undecidable from the text, or not decidable with the data used here. E5 was undetermined by its own rule and is not used as evidence here.

Status	Mechanism variant	Decisive evidence
Excluded	naive improvisation (Gaskell & Bower-type samples)	T3a, T3b
Excluded	stationary procedures (tables, grilles, automata, devices constant within a page)	T6a
Excluded	periodic devices of the tested design (period 5, cycle 4) with mixing weight ≥ 0.2	T4
Excluded	improvisation with smooth drift across pages	T5

Excluded	settings per line or paragraph as the source of the local dynamics	E1
Excluded	copy-and-modify with propagating innovations and vertical copying	T7b, T2c, T4b
Excluded	the tested natural-language controls, and systems that preserve lexical identity at the observed token boundaries	T3b, T7b
Excluded	the Naibbe cipher as published	T6a, E4; boundary MI, hapax share and page MI (Section 3.3)
Excluded	copying from a draft with different line breaks	T2a
Unresolved	explicit rule set with randomiser vs. practised habit	same generator describes both
Unresolved	message carried by the free choices (mimic-function encoding)	E7: not identifiable from text
Unresolved	direct composition vs. copying with identical line breaks	not separable by the transliteration
Open	a generator reproducing C6 and C7 together	best explicit generator 33/40

Read causally, which is an interpretation and not a measurement, the class describes a writer or procedure that:

- builds tokens in a strict slot order;
- makes each token's first glyph depend on the previous token's last glyph within a line;
- reuses recent forms, continuously across line and paragraph boundaries;
- varies its preferences among shared forms by page and quire, and drifts within a page about as fast as topic drift in prose;
- produces distinct forms at lines that end at the margin;
- seldom repeats a locally deviant form.

Each component of this reading was either required by a calibrated test or shown necessary by ablation on held-out pages. The causal mechanism behind each component (for example, whether the margin association arises from choosing the last word to fit the space) is not identified by these tests.

8. The message is not identifiable from the text (E7)

8.1 Construction

Every production process in the class makes free choices: at each glyph, several continuations are allowed with some probability. A message can be carried by making those choices depend on it.

The principle is not new. Wayner (1992) introduced mimic functions, and steganography theory shows that a message encoded to be indistinguishable from random bits can drive the random choices of a known channel without changing its output distribution (Cachin 2004; Hopper, Langford and von Ahn 2002). What E7 adds is a construction for a Voynich-like process on the manuscript's own layout, a measured capacity, and a test against the complete phenotype used throughout this paper. The rate-optimal way to do this is arithmetic decoding, a mimic function in the sense of Wayner (1992). The message bits are read as a binary fraction. At each choice the current interval is divided in proportion to the process's probabilities, and the sub-interval that contains the fraction is taken. A receiver who knows the process re-encodes the observed choices and obtains the bits.

We implemented this with the 32-bit coder of Witten, Neal and Cleary (1987), applied to the order-3 glyph automaton of the battery on the Voynich layout. Three conditions were compared, each through the full 40-statistic phenotype:

- random driving (3 seeds);

- compressed plaintexts: Latin, Italian and German, compressed with lzma;
- the same plaintexts as raw UTF-8 bits.

Proposition. Let a process emit a sequence of categorical choices with probabilities $p_t(\cdot \mid \text{history})$ known to sender and receiver. Let the choices be made by arithmetic decoding of a bit sequence U .

1. If U consists of independent, uniformly distributed bits, the output has the distribution of the process itself, up to the coder's finite precision. Hence every statistic of the text has the same distribution with and without a message.
2. A receiver who knows the p_t recovers U over the consumed length.
3. The number of bits carried equals the log-loss of the generated text under the process, whose expectation is the process's entropy.

Sketch. Arithmetic decoding partitions $[0, 1)$ into intervals whose lengths are the probabilities of the corresponding choice sequences. A uniformly distributed fraction falls into each with exactly that probability (Witten et al. 1987). Recovery follows because encoding is the inverse map on the consumed prefix. With a 32-bit range and the model counts used here, the per-choice probability error is far below the sampling noise of any statistic. Compressed data are close to independent uniform bits. The same principle underlies information-theoretic and complexity-theoretic definitions of secure steganography: a message encoded to be indistinguishable from uniform randomness can drive the random choices of a known channel without changing its distribution (Cachin 2004; Hopper, Langford and von Ahn 2002).

8.2 Results

All six message-driven runs returned their plaintext with 0 bit errors (Table 10). The process consumed about 352,812 bits per manuscript-sized output. Divided by the number of EVA characters generated (spaces excluded), this is 1.90 bits per glyph; divided by the number of tokens, 10.0 bits per token. The consumed bits correspond to about 44 kB (1 kB = 1,000 bytes) of lzma-compressed plaintext, roughly 145 thousand characters of Latin prose. These are capacities of this order-3 process on the manuscript's layout. They are not a measurement of how much information the manuscript carries.

For compressed plaintexts, 39 of 40 statistics agreed with random driving. The single disagreement (word-pair MI, three runs per condition) disappeared with eight runs per condition: 0.052 ± 0.011 against 0.053 ± 0.010 , Welch $p = 0.83$. Even raw, uncompressed plaintext bits agreed on 39 of 40 (T7b deviant recurrence differed).

Table 10. E7: message-driven output of the order-3 glyph process. Bits per glyph are bits consumed divided by the number of EVA characters generated, spaces excluded. Agreement counts are per condition group (3 plaintexts against 3 random seeds).

Condition	Bits consumed	Bits per glyph	Recovery mismatches	Statistics agreeing with random driving
compressed Latin	354,487	1.90	0	39/40
compressed Italian	352,347	1.91	0	39/40
compressed German	351,602	1.90	0	39/40
raw Latin	350,690	1.90	0	39/40
raw Italian	351,787	1.90	0	39/40
raw German	354,204	1.91	0	39/40

More runs and a richer process (R4). Three runs per condition give little power, and the order-3 automaton is the simplest member of the class. Two follow-up analyses were therefore pre-specified

in response to a review (Table 11, Figure 7).

- **E7 with 20 runs per condition.** 20 randomly driven runs were compared with 20 runs driven by lzma-compressed Latin, Italian, German and English at different offsets. After Holm correction, 0 of 40 statistics differ; 4 have unadjusted $p < 0.05$, where about 2 are expected by chance. All 20 message-driven runs returned their plaintext with 0 bit errors.
- **E8, a message through the E3 generator.** The same construction was applied to the best explicit generator of Section 6. Only choices the receiver can observe carry message bits: the glyphs of tokens built from the grammar, the choice of a reused recent form, and the choice of the final word of a line. Latent choices (page tilts, drift, the route between recency and grammar, slips) come from a key shared by sender and receiver. After Holm correction 0 of 40 statistics differ (4 unadjusted $p < 0.05$). All 20 runs returned their plaintext with 0 bit errors, at 1.49 bits per glyph, about 35 kB per manuscript.

Table 11. Message-driven against randomly driven output over the 40 statistics. Holm-significant: Welch tests (Fisher for T4) with Holm correction at $\alpha = 0.05$; about 2 unadjusted $p < 0.05$ are expected by chance among 40. The 20-run analyses and E8 were pre-specified in mechanism/REVISION_PLAN.md (R4). E8 carries the message only in choices the receiver can observe, so its capacity is lower.

Process	Message	Runs (message / random)	Bits per glyph	Recovery errors	Holm- significant	Unadjusted p < 0.05	Tolerance rule
order-3 automaton (E7)	compressed	3 / 3	1.90	0	–	–	39/40
order-3 automaton (E7, 20 runs)	compressed	20 / 20	1.90	0	0/40	4/40	40/40
order-3 automaton (E7, 20 runs)	raw text bits	20 / 20	1.90	0	0/40	3/40	40/40
E3 generator (E8)	compressed	20 / 20	1.49	0	0/40	4/40	40/40

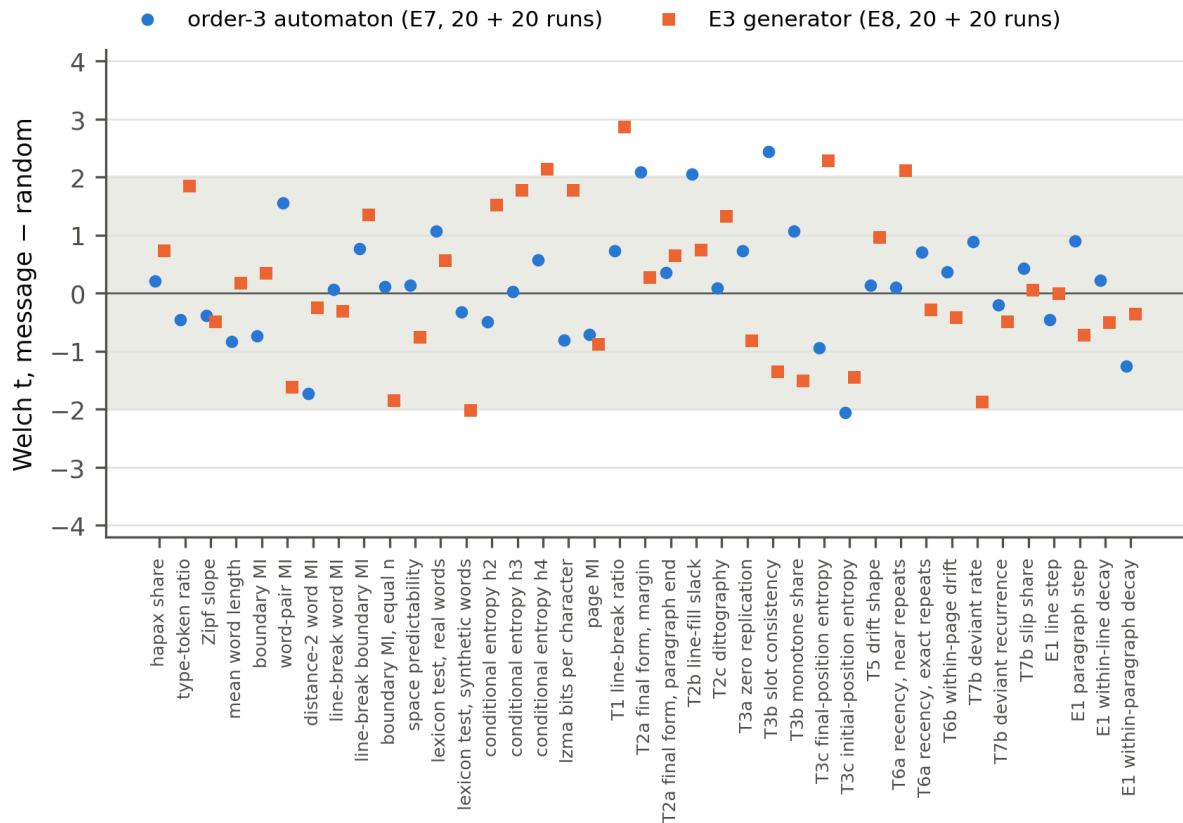


Figure 7. Welch t statistic of the difference between message-driven (lzma-compressed plaintext) and randomly driven output for each statistic, with 20 runs per condition, for the order-3 automaton (E7) and the E3 generator (E8). The shaded band is ± 2 ; no statistic differs after Holm correction.

8.3 What this does and does not show

E7 does not show that the Voynich text carries a message. It shows that the question cannot be settled from the text, for this class of encoders. Any process in the class can be driven by a message without changing the distribution of any statistic computable from the text. A message is identifiable only under a restriction on the encoder.

Within the restrictions tested here, the evidence is negative:

- systems that preserve lexical identity at the token boundaries produce recurring locally deviant forms, of which the text has very few;
- the Naibbe cipher lacks the recency, the boundary coupling and the page effects, also on its author's own ciphertext;
- the two tested letter-level ciphers remove the sublexical drift that the text keeps.

Arithmetic coding is a twentieth-century technique. An encoder of equivalent effect in the fifteenth century would be a historical claim requiring historical evidence. Untested historical encoders may exist.

The same argument covers the other unresolved alternative. A written rule set with a randomiser and a practised writer's habits are described by the same generator, and no text statistic separates them.

9. Discussion

9.1 What is new

Relative to earlier work, the contributions are these:

1. From the exploratory profile: the lexicon test; the compression residual against a drifting twin; a sufficient glyph process tested on held-out pages; the failure of ciphertext-feedback substitution and of state-dependent Naibbe variants, in which boundary coupling and hidden plaintext redundancy compete; the comparison of hand and section at fixed folio distance; and a text–illustration test with a calibrated sensitivity.
2. A pre-specified, calibrated separation of production hypotheses, with human gibberish as the improvisation control at the level of constraints and line structure. Gaskell and Bowern (2022) compared low-level statistics and noted that their samples were too short for page structure.
3. Line-final forms associated with the margin and much weaker at paragraph ends. Currier (1976) already described the line as a functional unit; the margin/paragraph contrast and its persistence on text-only pages are new.
4. Continuous recency across line and paragraph boundaries, which excludes settings that switch per line or paragraph. It is weaker than, and structurally different from, the copying of Timm and Schinner (2020): there are no propagating innovations and no vertical copying.
5. Very low recurrence of locally deviant forms, which separates the text from the language controls and from every generator we built, though not from the Naibbe cipher.
6. Page clustering that extends to the most frequent forms, with a rare-type gradient below every content control.
7. Held-out generator tests with ablations that turn "sufficient" into "necessary components plus two open specifics".
8. A Voynich-specific constructive non-identifiability result: a measured capacity for a Voynich-like process and a test against the full phenotype. The underlying principle is that of mimic functions and secure steganography (Section 8.1).

The rigidity measure quantifies the slot structure described by Stolfi (2000) and Zattera (2022). The boundary coupling and the predictability of spaces confirm Rozanova and Temerev (2026) with different estimators. The long-range structure reported by Montemurro and Zanette (2013) corresponds, in our measurements, to drift of the glyph statistics that follows the scribal hand more than the section (Section 3.4) and to the page effect of E5. The positional entropy collapse (T3c: Voynich 0.63, Naibbe 0.65, automaton 0.67, languages 0.78–0.91, gibberish 0.98) is reproduced by verbose ciphers, as Kinnison (2026) reported. It separates the text from natural language and naive gibberish, not procedure from cipher.

9.2 Why text statistics cannot settle the plaintext question

The debate between "meaningless" and "cipher" readings has often been conducted with statistics: entropy, Zipf's law, word lengths, keyword clustering. The proposition in Section 8 explains why no statistic of this kind can end it. A message compressed and carried by arithmetic decoding leaves the text distribution unchanged. The only statistics that can reveal a message are those that detect a departure of the text from the distribution of the message-free process, and such a test presupposes a model of the encoder.

Evidence for or against a message must therefore take one of three forms:

- tests against a specific, historically motivated encoder class, as done here for systems that preserve lexical identity and for the Naibbe cipher. Other historical systems need tests of their own. The Polygraphia III (Hermes 2022) is an obvious candidate: it encodes each plaintext letter as one invented token, and its tables and a simulation are public. Hermes compared it on entropy, word lengths and near-identical neighbours; whether it can also

produce the boundary dependency (C2), the recency (C4) and the page variation (C5) found here is untested. It is also about a century younger than the manuscript;

- external information, such as a key, a bilingual passage, or a demonstrable link between text and image, of which Section 3.4 found none at the tested sensitivity;
- physical and historical evidence about how the manuscript was made.

This is not special to the Voynich manuscript. It is a property of every text produced by a stochastic process with free choices. It extends the conclusion of Bower and Gaskell (2022), that unusual character-level statistics are no evidence that the text is gibberish: no statistic computed from the text can decide the question without a model of the encoder.

9.3 Rule, habit or message

Within the class, three considerations weigh without deciding.

- **No periodicity of the tested kind.** There is no device footprint at the power we could test, for the one device design we tested.
- **Human production plausibility.** Recency, page and quire variation, margin-associated endings and drift are what one expects of a human writer, and a rule set would have to specify each explicitly, including a continuously drifting preference. Our own human control is too short to show recency or drift, so this consideration rests on general grounds, not on our data.
- **No tested cipher fits.** Every historical-style cipher we tested leaves traces that the text lacks.

The most parsimonious reading is writers composing on the page within a strict learned grammar, choosing forms under short-term recency and page- and quire-dependent preferences, and choosing each line's last word with the margin in view. We label this a parsimony judgement, not a finding. Whether a message rode on those choices remains undecidable from the text.

9.4 Transmission

Copying from a re-lined draft is excluded (T2a). No copying signature was found: no dittography, and no word-level corrections among the annotated ones. The final forms depend on whether a line ends at the margin. The simplest interpretation is that the last token was chosen knowing where that line ends, that is, composition on the page. Copying from a draft with identical lines and margins cannot be excluded from the transliteration.

9.5 Red-team checks

Table 12. Red-team checks. The first nine were run before writing and are post hoc. The last four respond to a review: R1 is post hoc; R2–R4 were pre-specified in mechanism/REVISION_PLAN.md before computation.

Claim	Attack	Check	Result
C3 margin-associated endings	margin and paragraph effects not actually different	page bootstrap of the difference	0.046 (0.026, 0.062)
C3	lines end at drawings, not margins	text-only stars section	margin 0.156 vs paragraph 0.033 (n = 872, 291)
C1 rigidity	artefact of multigraph segmentation	raw EVA characters	Voynich 0.034 / 0.856 vs gibberish 0.098 / 0.679
C1 hardness	depends on sample size	full-size zero replication	Voynich 0.0043 vs languages 0.0017–0.0042: no separation at full size

C1, C3, C4	driven by one scribe	per hand (1, 2, 3)	T3a 0.036–0.049; T3b 0.85–0.89; T6a 0.14–0.17; T2a 0.064–0.155
C7 low recurrence of deviant forms	depends on deviance definition	page-level blocks	Voynich 0.5% vs languages 2.1%–5.1%, generator 2.3%, Naibbe 0.4%
C7	depends on double coding or on one hand	no double coding; per hand, whole-text reference	1.7%; hands 1.1%–2.5%
E7	single disagreement is a real trace	8 runs per condition	word-pair MI 0.052 ± 0.011 vs 0.053 ± 0.010 , $p = 0.83$
All	transliteration choice	Takahashi transliteration	T3a, T3b, T5, T6a, T7b, T2a, E1, E5 replicate
C1 rigidity	mean over writers hides writers as rigid as the Voynich text	each writer against Voynich stretches of the same length (R1)	36 of 36 writers below the 5th percentile; writers 0.57–0.78
Naibbe exclusion	the reimplementation differs from the published cipher	author's own ciphertext (R2, R3)	mapping 100.0% valid, 7/8 statistics agree; author's output: T6a 0.000, boundary MI 0.0010, page MI 0.000
E7	three runs per condition are too few	20 runs per condition (R4a)	Holm-significant 0/40; unadjusted $p < 0.05$ 4/40
E7	only the simplest process was tested	message through the E3 generator (E8, R4b)	Holm-significant 0/40; recovery errors 0

Two checks changed the paper. The hardness contrast (T3a) holds at matched short-stretch size but not at full size, so C1 rests mainly on rigidity. The very low recurrence of deviant forms is shared by the Naibbe cipher, so it excludes systems that preserve lexical identity, not homophonic ones. The four checks prompted by the review left the claims unchanged: the rigidity contrast holds for every gibberish writer, the Naibbe reimplementation matches the author's own output and the exclusion holds on that output, and the non-identifiability result holds with 20 runs per condition and for the best generator.

10. Limitations

- **Blindness is procedural, not epistemic.** The author knew many Voynich properties before designing the tests, and both halves' values were known before E4–E6. The analysis plans are timestamped by commits in a repository that was private until the work was complete; there was no external registration.
- **The profile is exploratory.** Section 3 was not specified in advance. Its table-and-grille and copy-and-modify generators are reconstructions of published descriptions that could not be checked against their authors' outputs; the Naibbe reimplementation was checked against its author's published ciphertext (R2). The illustration test uses crude colour features of folios 1–58 only, and its sensitivity (about a tenth of a page's tokens) bounds what it can exclude.
- **Two transliterations, one source.** ZL3b and IT2a are independent readings of the same images; transliteration conventions (spaces, ligatures) remain a shared source of error.
- **Controls.**
 - The human gibberish is short, modern and in the Latin alphabet; no data exist on long-practised improvisation in an invented script, which is the main reason the executor question stays open.
 - The language controls are modern editions, and none is a heavily abbreviated medieval text.
 - Most generator controls were fitted to the Voynich text.

- The Vulgate control was lost to an empty file.
- **Estimator dependence.**
 - The line-break ratio ranges from about 0.03 (pooled estimator) to 0.22 (page-conditional estimator, the only one valid under writer heterogeneity).
 - The hardness contrast depends on sample size.
 - T4's power is known only for the tested device design (period 5, cycle 4).
- **Page order** is the binding order, which departs from the writing order in places. This weakens T5 and T4c.
- **Corrections** rest on 24 annotations made for reading purposes; line fill uses glyph counts, not physical widths.
- **Sufficiency of the class** is not demonstrated: the best generator reproduces 33 of 40 statistics.
- **E7 and E8** check the proposition for two processes, the order-3 automaton and the E3 generator. The proposition covers any process with known choice probabilities; E8 carries the message only in observable choices, so its capacity is a lower bound for that process.
- **The 40 statistics are correlated** and assessed with a tolerance rule, not a joint test.
- **Analysis changes after calibration.** Several estimators were changed after their calibration failed: the T1 null model, the E1 expectations and the E5 intervals (Table S27). Every change was made before the affected Voynich statistic was computed. The T1 terms were computed once on the Voynich text during the diagnosis.
- **Undetermined results.** E5 was undetermined by its pre-specified rule, and E4 did not separate content drift from habit drift. The paper uses them only descriptively and not as evidence for any exclusion.
- **Tokens and words.** Whether Voynich tokens are words is itself open (Section 3.2). Exclusions phrased in terms of lexical identity refer to identity at the transcribed token boundaries.

11. Conclusion

An exploratory profile shows that a local, low-order glyph process with drifting parameters suffices for the measured statistics of the Voynich text, and that neither a lexicon nor information about the colours of the drawings is measurable beyond it. Pre-specified, calibrated tests then narrow the production of the text to a class defined by nine measured properties. The text excludes:

- naive improvisation;
- stationary tables and grilles;
- periodic devices of the tested design;
- settings that switch per line or paragraph;
- copy-and-modify with propagating innovations;
- the tested natural-language controls, systems that preserve lexical identity at the token boundaries, and the Naibbe cipher as published;
- copying from a re-lined draft.

It leaves three alternatives that it cannot decide between: an explicit rule set, a practised writer's habits, and a message carried by the free choices. For the last, we showed constructively that a message can be present without leaving any trace in the statistics used here. For uniformly distributed message bits, it leaves a trace in no statistic that can be computed from the text.

A set of production constraints can be recovered from the text, but whether the free choices of the

constrained process carry a message is not identifiable from the text alone. Progress beyond this point requires evidence the text does not contain:

- long-term improvisation and rule-execution experiments with human writers;
- physical examination of the manuscript for session boundaries and corrections;
- historical sources describing a procedure of this kind.

Data and code availability

All code, analysis plans, deviation logs, results and the build scripts for this paper are at <https://github.com/beyse/voynich>. This version of the paper was built from commit a0b0b86 of that repository. The profile analysis is in `voynich/` and `scripts/` with its logs in `results/logs/`; the pre-specified experiments are in `mechanism/` (analysis plans in the files `*_PREREG.md`) with their results in `results/mechanism/`; the paper sources are in `paper/`. The transliterations are those of voynich.nu (Zandbergen and Landini 2025; Takahashi 1999). The gibberish corpus is that of Gaskell and Bower (2022), under its licence, which requires citation. The Naibbe tables are those of Greshko (2025), under the licence of that repository, which requires attribution; his example ciphertext used in R2 and R3 is fetched from his repository at a fixed commit by `mechanism/revision.py`. The revision analyses are in `mechanism/revision.py`, `e7b.py` and `e8.py`, with results in `results/mechanism/revision/`, `e7b/` and `e8/`. The reference texts (Project Gutenberg, The Latin Library, Sefaria) and the Beinecke facsimile are not redistributed; the repository documents how to obtain them.

Licence

This preprint, its supplement and figures, and the code, analysis plans and results in the accompanying repository are dedicated to the public domain under the Creative Commons CC0 1.0 Universal dedication (<https://creativecommons.org/publicdomain/zero/1.0/>). Third-party data in the repository keep their own licences, as stated above.

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Disclosure of the use of generative AI

This work was carried out with Claude Opus 5.5 (Anthropic; model identifier `claude-opus-5-5`), used through Claude Code. How it was used:

- The author set the research questions and the scope, fixed the claims policy (Section 4.2), and decided between alternatives at each stage.
- Under the author's direction, the AI system designed and wrote the analysis code, ran the analyses, drafted the analysis plans, and drafted the text, tables and figures of this paper and its supplement.

It was used because the author worked alone and without funding, and the tool made an analysis of this scope feasible. Every number in the paper is computed from the result files by the build script in the public repository and can be reproduced from it. The author takes full responsibility for the content. The paper has not been peer reviewed.

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References

- Bowerman, C. L., and D. E. Gaskell. 2022. Enciphered after all? Word-level text metrics are compatible with some types of encipherment. *CEUR Workshop Proceedings* 3313, International Conference on the Voynich Manuscript 2022. <https://ceur-ws.org/Vol-3313/paper6.pdf>.
- Bowerman, C. L., and L. Lindemann. 2021. The linguistics of the Voynich manuscript. *Annual Review of Linguistics* 7:285–308. doi:10.1146/annurev-linguistics-011619-030613.
- Cachin, C. 2004. An information-theoretic model for steganography. *Information and Computation* 192 (1):41–56.
- Church, K. W., and W. A. Gale. 1995. Poisson mixtures. *Natural Language Engineering* 1 (2):163–190.
- Currier, P. H. 1976. Papers on the Voynich manuscript. In *New research on the Voynich manuscript: Proceedings of a seminar*, ed. M. E. D'Imperio. Washington, DC. Reproduced at https://www.voynich.nu/extra/curr_main.html.
- D'Imperio, M. E. 1978. *The Voynich manuscript: An elegant enigma*. Fort George G. Meade, MD: National Security Agency.
- Davis, L. F. 2020. How many glyphs and how many scribes? Digital paleography and the Voynich manuscript. *Manuscript Studies* 5 (1):164–180. doi:10.1353/mns.2020.0011.
- Gaskell, D. E., and C. L. Bowerman. 2022. Gibberish after all? Voynichese is statistically similar to human-produced samples of meaningless text. *CEUR Workshop Proceedings* 3313, International Conference on the Voynich Manuscript 2022. Data: <https://github.com/danielgaskell/voynich>.
- Greshko, M. A. 2025. The Naibbe cipher: A substitution cipher that encrypts Latin and Italian as Voynich Manuscript-like ciphertext. *Cryptologia*. doi:10.1080/01611194.2025.2566408. Code and tables: <https://github.com/greshko/naibbe-cipher>.
- Hermes, J. 2022. Polygraphia III: The cipher that pretends to be an artificial language. *CEUR Workshop Proceedings* 3313, International Conference on the Voynich Manuscript 2022. <https://ceur-ws.org/Vol-3313/paper7.pdf>.
- Hopper, N. J., J. Langford, and L. von Ahn. 2002. Provably secure steganography. In *Advances in Cryptology – CRYPTO 2002*, LNCS 2442, 77–92. Berlin: Springer.
- Kinnison, R. 2026. The Voynich Manuscript's positional entropy collapse: Comparative evidence from languages and ciphers. *Cryptologia*. doi:10.1080/01611194.2026.2697318.
- Lindemann, L., and C. Bowerman. 2020. Character entropy in modern and historical texts: Comparison metrics for an undeciphered manuscript. arXiv:2010.14697.
- Mantel, N. 1967. The detection of disease clustering and a generalized regression approach. *Cancer Research* 27 (2):209–220.
- Montemurro, M. A., and D. H. Zanette. 2013. Keywords and co-occurrence patterns in the Voynich manuscript: An information-theoretic analysis. *PLoS ONE* 8 (6):e66344. doi:10.1371/journal.pone.0066344.
- Paninski, L. 2003. Estimation of entropy and mutual information. *Neural Computation* 15 (6):1191–1253.
- Rozanova, L., and A. Temerev. 2026. A glyph is not a letter, a token is not a word, a space is not a space: What the units of Voynichese are not. arXiv:2608.17096.
- Rugg, G. 2004. An elegant hoax? A possible solution to the Voynich manuscript. *Cryptologia* 28 (1):31–46. doi:10.1080/0161-110491892755.
- Sennrich, R., B. Haddow, and A. Birch. 2016. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*, 1715–1725. Berlin: Association for Computational Linguistics.
- Stolfi, J. 2000. A grammar for Voynichese words. Mirrored at <https://www.voynich.nu/hist/stolfi/grammar.html>.
- Takahashi, T. 1999. Voynich manuscript transcription. Distributed as IT2a-n.txt at <https://www.voynich.nu/>

transcr.html.

Timm, T., and A. Schinner. 2020. A possible generating algorithm of the Voynich manuscript. *Cryptologia* 44 (1):1–19. doi:10.1080/01611194.2019.1596999.

Wayner, P. 1992. Mimic functions. *Cryptologia* 16 (3):193–214. doi:10.1080/0161-119291866883.

Witten, I. H., R. M. Neal, and J. G. Cleary. 1987. Arithmetic coding for data compression. *Communications of the ACM* 30 (6):520–540.

Zandbergen, R. n.d. The radio-carbon dating of the Voynich MS. <https://www.voynich.nu/extra/carbon.html>. Accessed 23 September 2026.

Zandbergen, R., and G. Landini. 2025. ZL transliteration of the Voynich manuscript, version 3b (IVTFF format, extended EVA). <https://www.voynich.nu/transcr.html>.

Zattera, M. 2022. A new transliteration alphabet brings new evidence of word structure and multiple "languages" in the Voynich manuscript. International Conference on the Voynich Manuscript 2022, University of Malta.

References cited only in the supplement are included in this list.